30-Day ML & AI Mastery Plan

Prepared for Shubham · 28 October 2026

How this course works

In 30 days at 2–5 hrs/day (about 60–150 hours) you will reach solid working proficiency across the whole curriculum and build one portfolio-grade capstone; true mastery comes from repeating these projects over the following months. Days 1–30 teach every topic in the curriculum, each ending with a runnable demo; the capstone is built from scratch in Days 31–45.

Daily format — say "Day N" in chat and I teach that day's lesson:

1. **Concept (30–45 min)** — intuition first, then the math, with diagrams.
2. **Live demo (45–60 min)** — a complete Colab notebook on a real dataset you run cell by cell.
3. **Exercise (30–60 min)** — you modify the demo; I review your code.
4. **Stretch (optional, +1–2 hrs on 5-hr days)** — a deeper variant or paper reading.
5. **Quiz (10 min)** — 5 questions to lock it in.

Setup (do once, ~30 min)

- [ ] Google account → Google Colab (colab.research.google.com) — free GPU for Weeks 3–4
- [ ] Local: Python 3.11 + VS Code + `pip install numpy pandas matplotlib seaborn scikit-learn scipy jupyter`
- [ ] Kaggle account (datasets + free GPU notebooks)
- [ ] Hugging Face account (needed from Day 26)
- [ ] GitHub repo `ml-30-days` — commit every day's notebook (this becomes your portfolio)
- [ ] An LLM API key (OpenAI or Groq free tier) before Day 27

On 2-hour days do steps 1–3 only; **on 5-hour days** add the stretch and start capstone EDA early from Day 15 onward.

Your edge: your C#/.NET, SQL Server and React background maps directly onto Pandas (≈ LINQ + SQL), FastAPI (≈ ASP.NET Core Web API) and the capstone dashboard. I'll draw those parallels as we go.

Week 1 (Days 1–7): Bridge module — Python, math and data foundations

Week 1 gives you the Python, probability, linear algebra and calculus that every later algorithm is built on, coded rather than memorised.

Day	Topic	Key concepts	Live demo
1	Python proficiency	Syntax, data types, lists/tuples/dicts/sets, comprehensions, functions, *args/**kwargs, lambdas, classes, modules	Student-marks analyzer using only core Python
2	Key libraries	NumPy arrays & broadcasting, Pandas DataFrames, groupby/merge, Matplotlib/Seaborn plots, SciPy overview	Full EDA of the Titanic dataset
3	Probability	Events: mutually exclusive vs non-exclusive, independent vs dependent; conditional probability, Bayes' theorem; discrete vs continuous random variables; PMF, PDF, CDF; Bernoulli, Binomial, Poisson, Normal	Simulating dice/coins; a Bayes spam filter by hand
4	Statistics	Mean/median/mode, variance, std, IQR, skew; normalization vs standardization; sampling, Central Limit Theorem, confidence intervals, hypothesis tests	Standardizing real features; CLT simulation
5	Linear algebra	Vectors, L1/L2/∞ norms, dot product, matrices and operations, inverse, rank, linear dependence/independence, linear transformations, eigenvalues/eigenvectors	Rotating, scaling and shearing an image with matrices
6	Calculus	Derivatives, chain rule, partial derivatives, gradients, integration (area under PDF), least-squares error	Gradient descent from scratch fitting a line
7	Overview of AI/ML + data	AI vs ML vs DL, supervised/unsupervised/reinforcement, formulating real-world problems for ML, representing real-world data (tabular, text, image, audio), data processing & visualization, web scraping	Scrape a website → clean → visualize; frame 3 business problems as ML tasks

Week 1 checkpoint: a notebook that loads a raw CSV, cleans it, standardizes it and fits a line with your own gradient descent.

Week 2 (Days 8–14): Classical machine learning

Week 2 covers the scikit-learn toolkit end to end: you'll train, tune and honestly evaluate every classical algorithm in the curriculum, which is what most industry ML still runs on.

Day	Topic	Key concepts	Live demo

8	Principles of ML	Train/validation/test split, cross-validation, overfitting vs generalization, bias–variance, k-NN classifier, linear classifier, decision boundaries	k-NN on Iris and Wine; plotting boundaries as k changes
9	Regression & classification	Linear regression (normal equation vs GD), Ridge/Lasso, logistic regression, performance metrics: MAE, RMSE, R^2, confusion matrix, precision, recall, F1, ROC-AUC, PR curves	House-price regression; bank term-deposit classifier
10	Features	Feature extraction, normalization, transformation (log, Box-Cox), encoding, missing values, imbalance (SMOTE, class weights), feature selection (filter, wrapper, embedded), sklearn Pipelines	Leak-free pipeline that lifts a baseline's F1
11	Ensembles	Decision trees (Gini, entropy), sampling: bootstrap & bagging, bagging classifier, voting classifier, Random Forest, AdaBoost, Gradient Boosting, XGBoost, feature importance, SHAP	Employee attrition prediction, 6 models compared
12	SVM & MLP	Separability vs non-separability, max-margin, soft margin, kernel trick (linear, poly, RBF), MLP classifier & regressor, hyperparameter tuning (GridSearch, Optuna)	SVM kernels on non-linear data; sklearn MLP vs SVM
13	Unsupervised learning	k-Means, elbow & silhouette, hierarchical clustering & dendrograms, DBSCAN, Gaussian Mixtures	RFM customer segmentation for retail marketing
14	Dimensionality reduction	PCA (from eigenvectors), simple PCA, PCA EigenFaces, manifold learning, ISOMAP, t-SNE, UMAP	EigenFaces face reconstruction; t-SNE of MNIST digits

Week 2 checkpoint: an end-to-end Kaggle-style notebook (EDA → pipeline → 5 models → tuned ensemble → metrics report) on the bank-marketing dataset.

Week 3 (Days 15–21): NLP foundations, neural networks and computer vision

Week 3 moves from hand-built features to learned ones: you'll write backpropagation yourself once, then use PyTorch and Keras to build image models that run on video. Switch Colab to a GPU runtime from Day 17.

Day	Topic	Key concepts	Live demo
15	Representing text & language	NLTK: tokenization, stemming, lemmatization, stopwords, POS, NER; Bag-of-Words, n-grams, TF-IDF, word2vec/GloVe embeddings; text classification, text generation (n-gram), sentiment analysis	Tweet sentiment classifier with TF-IDF + logistic regression
16	Neural networks from scratch	Perceptron, why XOR needs non-linear solutions, multi-layer perceptron, activations, loss functions, gradient descent (batch, SGD, mini-batch, momentum, Adam), backpropagation	2-layer network in pure NumPy solving XOR and moons
17	PyTorch, TensorFlow & Keras	Tensors, autograd, Dataset/DataLoader, training loop; Keras Sequential & functional API; dropout, batch norm, early stopping, weight decay to avoid overfitting	Same MNIST model in PyTorch and Keras, side by side
18	CNNs	Convolution, filters, stride, padding, pooling, feature maps, appreciating CNNs (what each layer sees), features for perception, image augmentation	CIFAR-10 CNN + filter/activation visualisation
19	Beyond AlexNet & transfer learning	AlexNet → VGG → Inception → ResNet → EfficientNet → Vision Transformer; transfer learning vs fine-tuning, freezing layers, learning-rate schedules	Food image classifier with ResNet50; emotion recognition from faces (FER-2013)
20	Computer vision	Image classification vs object detection (YOLO, IoU, NMS, mAP) vs segmentation (semantic: U-Net; instance: Mask R-CNN; SAM), OpenCV, video processing	YOLO detecting objects in a live webcam/video; U-Net food segmentation
21	Autoencoders & Siamese networks	Autoencoders, denoising AE, reconstruction of images, latent space, anomaly detection via reconstruction error, variational AE, Siamese networks, contrastive & triplet loss	Denoising MNIST; bearing-vibration anomaly detector; face-similarity Siamese net

Week 3 checkpoint: a fine-tuned image classifier above 85% accuracy, with Grad-CAM heat-maps showing what it looks at.

Days 22–30: Sequences, generative AI, transformers, LLMs, RAG, agents and deployment

The final nine days take you from sequence models to shipping an LLM-powered app behind a FastAPI endpoint — the stack most AI job descriptions ask for in 2026.

Day	Topic	Key concepts	Live demo
22	RNN, LSTM & GRU	Sequence modelling, vanishing gradients, LSTM gates, GRU, bidirectional RNNs, time-series applications: windowing, stationarity, ARIMA baseline vs LSTM, walk-forward validation	Stock price and daily logistics demand forecasting

23	Speech & ML in speech/NLP	Audio as signals, spectrograms, MFCCs, features for speech perception, speech emotion recognition, speech-to-text with Whisper	Emotion recognition from speech (RAVDESS); Whisper transcription
24	GANs & generative models	Generator vs discriminator, DCGAN, mode collapse, conditional GAN, SRGAN for image super-resolution, intro to diffusion models (Stable Diffusion)	DCGAN generating faces/digits; SRGAN upscaling
25	Attention & Transformers	Seq2seq + attention, self-attention, multi-head attention, positional encoding, encoder vs decoder; BERT, GPT, T5 architectures	Build a mini-transformer in PyTorch; visualise attention
26	Hugging Face & fine-tuning	Transformers library, pipelines, tokenizers, Datasets, Trainer; fine-tuning BERT (classification, Quora question pairs), fine-tuning GPT-2; LoRA/PEFT, QLoRA	Fine-tune BERT for sentiment; LoRA on a small GPT
27	LLMs, RAG & LangChain	OpenAI API (ChatGPT), Llama via Ollama, prompt engineering, embeddings, vector databases (FAISS/Chroma), chunking, retrieval, LangChain RAG chains, evaluation	Chat-with-your-PDFs RAG app
28	Agentic AI & human-in-the-loop	Tool/function calling, ReAct, LangGraph agents, memory, multi-agent patterns, MCP, human-in-the-loop approvals, guardrails	Research agent that searches, summarises and asks for approval before acting
29	Deployment & practical issues	FastAPI model serving, Docker, Streamlit/Gradio UIs, model compression (quantization, pruning, knowledge distillation, ONNX), data drift, monitoring, MLflow, deployment on Azure / Hugging Face Spaces	Deploy Day 26's BERT model as a FastAPI + Docker API; quantize it 4× smaller
30	Recommenders, optimal routes & capstone kickoff	Content-based vs collaborative filtering, matrix factorization, neural CF; finding the optimal route (Dijkstra, A*, vehicle routing with OR-Tools); capstone selection and problem statement	MovieLens recommender; delivery-route optimiser

Industry lectures (weekends, ~1 hr each): Andrej Karpathy's *Neural Networks: Zero to Hero*, 3Blue1Brown's neural-network series, and Stanford CS231n / CS224n lectures on the topics you just covered.

Capstone (Days 31–45): Real-time Tweet Analytics with an AI insights assistant

Recommended pick: **Real-time system for Tweet Analytics**, because it alone exercises classical ML, deep learning, transformers, RAG, agents and full-stack deployment — and your React/.NET/Azure experience makes the live dashboard a standout in the IIIT Hyderabad demo. Final choice is made on Day 30; the shortlist below are strong alternatives.

What it does: ingests a live stream of posts, classifies sentiment and emotion, extracts topics and entities, detects trending spikes, and lets a user ask questions like "Why did sentiment on #budget drop this afternoon?" answered by a RAG assistant over the stream.

Pipeline

1. **Ingest** — X's API is paid, so stream either the free Bluesky firehose or a replayed Kaggle tweet dataset (Sentiment140 / TweetEval) through a Python producer → queue.
2. **Process** — clean with NLTK/regex; baseline TF-IDF + logistic regression vs fine-tuned BERTweet/RoBERTa for sentiment and emotion; spaCy NER; BERTopic clustering.
3. **Detect** — rolling-window volume and sentiment; anomaly detection flags trending spikes.
4. **Store** — SQLite/Postgres for metrics; Chroma vector DB for post embeddings.
5. **Ask** — LangChain RAG assistant (Llama or OpenAI) that retrieves relevant posts and explains trends, with an agent that can query metrics.
6. **Serve** — FastAPI backend + React (or Streamlit) live dashboard, Dockerised, deployed to Azure or Hugging Face Spaces.

Build timeline (2–3 hrs/day)

Days	Phase	Output
31–32	Problem statement, data source, EDA	Proposal doc + EDA notebook
33–35	Baselines and BERT fine-tuning	Model comparison table (accuracy, F1, latency)
36–37	Streaming pipeline + topic/trend detection	Working end-to-end stream
38–40	RAG assistant + FastAPI	Q&A endpoint with cited posts
41–42	Dashboard, Docker, deployment	Live public URL
43–45	Report, slides, demo video, rehearsal	Submission package

Submission package

- [ ] GitHub repo with README (setup, architecture diagram, results, screenshots)
- [ ] Project report: problem, data, EDA, methods, experiments, results, limitations, ethics, future work
- [ ] 12–15 slide presentation for the IIIT Hyderabad panel
- [ ] 3-minute demo video + live demo
- [ ] Viva prep: 30 likely panel questions with answers

Strong alternatives

Project	What it showcases	Difficulty
Quora question pairs with same intent	Siamese networks, BERT fine-tuning, embeddings	Medium
Talent retention & attrition prediction	Tabular ML, XGBoost, SHAP explainability + an HR GenAI assistant	Low–medium (safest)
Food image segmentation	U-Net, SAM, transfer learning, augmentation	Medium–high
Image super-resolution with GAN	SRGAN, perceptual loss, PSNR/SSIM	High (GPU-heavy)
Daily demand forecasting for logistics	Time series, LSTM vs gradient boosting, route optimisation	Medium

Curriculum coverage map

Every item in your IIIT Hyderabad syllabus is mapped to a day below; extras I added are listed last.

Syllabus area	Items	Day(s)
Bridge: Python	Basics, data types & structures, functions, library packages	1–2
Bridge: Probability & statistics	Continuous/discrete RVs, mutually exclusive & independent events, PDF, CDF, centre & spread, normalization & standardization	3–4
Bridge: Calculus & linear algebra	Vectors, norms, matrices & operations, linear (in)dependence, linear transformations, least-square error, derivatives, chain rule, integration, gradients	5–6
Module 1	NumPy/SciPy/Pandas/Matplotlib, AI/ML overview, principles of ML, data processing & visualization, feature selection & extraction, simple algorithms, classification & regression, dimensionality reduction, text representation, bootstrap & bagging, problem formulation, real-world data, manifold & unsupervised learning, overfitting & generalization, performance metrics	2, 7–15
Module 2	Problem formulation, non-linear solutions & MLP, gradient descent & backprop, PCA EigenFaces, SVM & kernels, NLP, features for perception, ensembles (RF, AdaBoost)	7, 11–16, 18, 23
Module 3	Intro to DL, PyTorch/TensorFlow/Keras, CNNs, autoencoders, RNN/LSTM/GRU, time series, recommendation systems, beyond AlexNet & transfer learning, FastAPI, deployment, model compression, Siamese networks & GANs, computer vision, speech & NLP, attention & transformers (BERT, GPT, T5), Hugging Face, fine-tuning BERT & GPT, LLMs (OpenAI API, Llama), RAG & LangChain, human-in-the-loop & agentic AI	16–30
Module 4 / experiential	Web scraping & NLTK, k-NN, linear classifier, decision tree, bagging & voting classifiers, linear & logistic regression, PCA, ISOMAP, t-SNE, k-Means & hierarchical, XGBoost, reconstruction of images, sentiment analysis, emotion recognition from face & speech, optimal route, image augmentation/segmentation/classification, object detection, video processing	7–15, 18–23, 30
Extras added	Cross-validation & tuning (Optuna), SHAP explainability, class imbalance, Grad-CAM, Vision Transformers, diffusion models, LoRA/QLoRA, vector databases, LangGraph & MCP, Docker, MLflow, data drift	Throughout